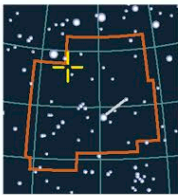


SC SPIRAL AND IRREGULAR GALAXIES

Whirlpool Galaxy



**CATALOG NUMBERS** M51,  
NGC 5194,  
NGC 5195

**DISTANCE**  
31 million light-years

**DIAMETER**  
100,000 light-years

**MAGNITUDE** 8.4

CANES VENATICI

Discovered by Charles Messier (see p.73) in 1773, the Whirlpool Galaxy is now known to be a pair of galaxies that is interacting—the brightest and clearest example of such a pair visible from Earth. The individual components are a spiral galaxy viewed face-on (NGC 5194) and a smaller irregular galaxy (NGC 5195). In visible light, the connection between them cannot be seen, but images at other wavelengths reveal an envelope of gas connecting the two. One effect of the interaction is to enhance the density wave in the larger galaxy, triggering increased star formation and making its spiral arms stand out very clearly. The Whirlpool was in fact the first “nebula” in which spiral structure was recognized, by William Parsons (see panel, right).

The interaction has also triggered increased activity in the cores of both of the galaxies—NGC 5195 is



CONTRASTING PAIR

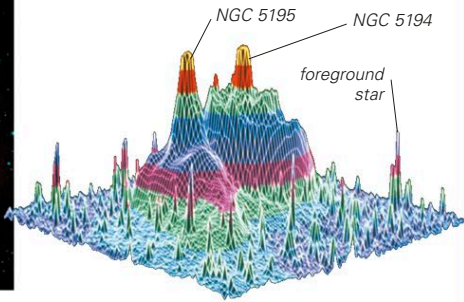
This infrared image, taken by the Spitzer Space Telescope, shows the Whirlpool Galaxy and its companion. The Whirlpool itself is rich in dust, which is colored red, while the companion is largely dust-free and appears blue.

undergoing a burst of star formation, which explains its unusual brightness, while NGC 5194’s core is also much brighter than expected. It is even classified by some astronomers as an active Seyfert galaxy (see p.320).

The Whirlpool Galaxy is very bright despite its distance, indicating that it is

LIGHT INTENSITY

Plotting the intensity of light from different regions of M51 reveals the brightness of the two galactic cores (the twin peaks on the graph).



large and luminous—it is similar in size to the Milky Way but brighter overall because of the large young star clusters in its spiral arms. It is thought to be the dominant member of a small group of galaxies, called simply the M51 group, which also includes the galaxy M63.

WILLIAM PARSONS

William Parsons (1800–1867) was an Irish nobleman who used his great wealth to build the largest telescope of his time and made the first detailed studies of nebulae. In 1845, he made detailed drawings and noticed the spiral structure of some “nebulae,” as galaxies were thought to be at the time. This was an important step to discovering that galaxies were not nebulae but separate star systems.



PARSONS’S SKETCH OF M51



LUMINOUS WHIRLPOOL

This Hubble image combines data from different filters to reveal detail in M51, such as dark dust behind each spiral arm and bright pink regions of star birth.